



Introduction to Brain and Cognition

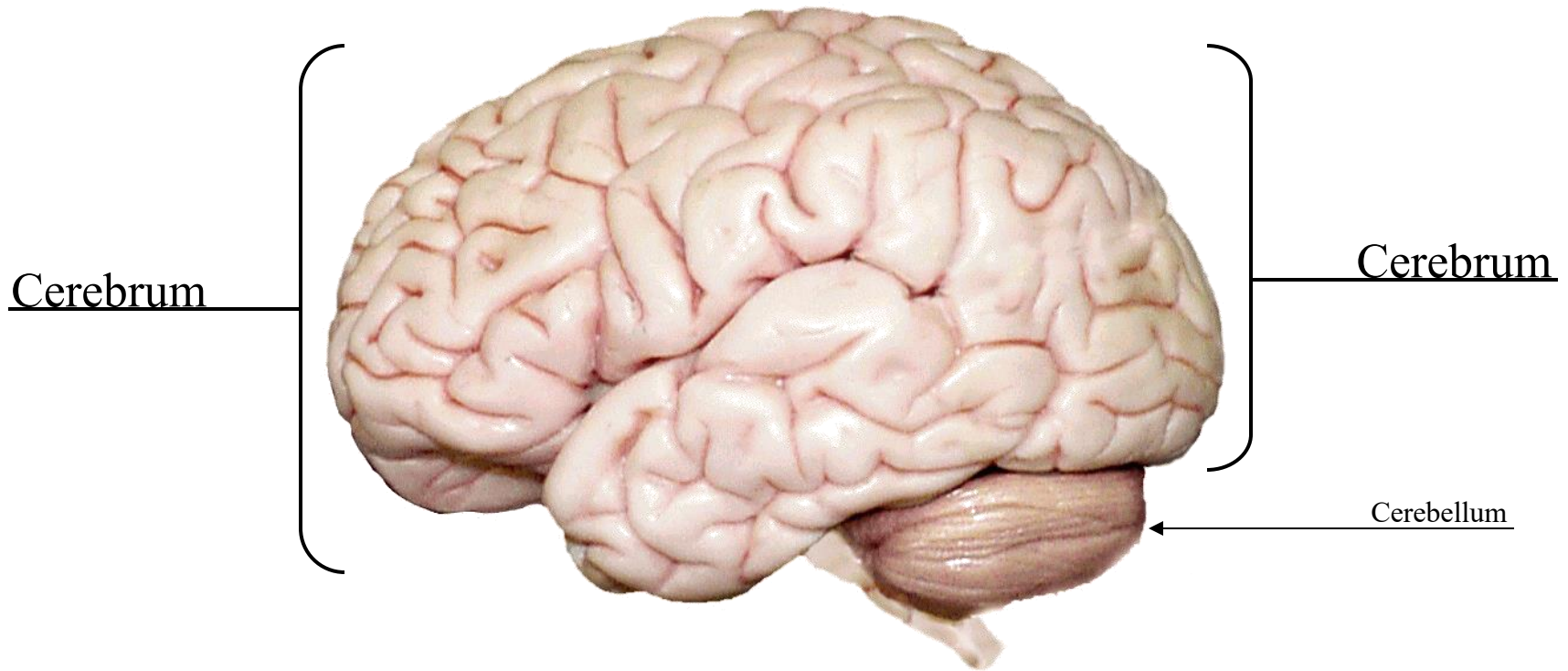
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READING ASSIGNMENT: Chapter 2, Reisberg (2019). Cognition

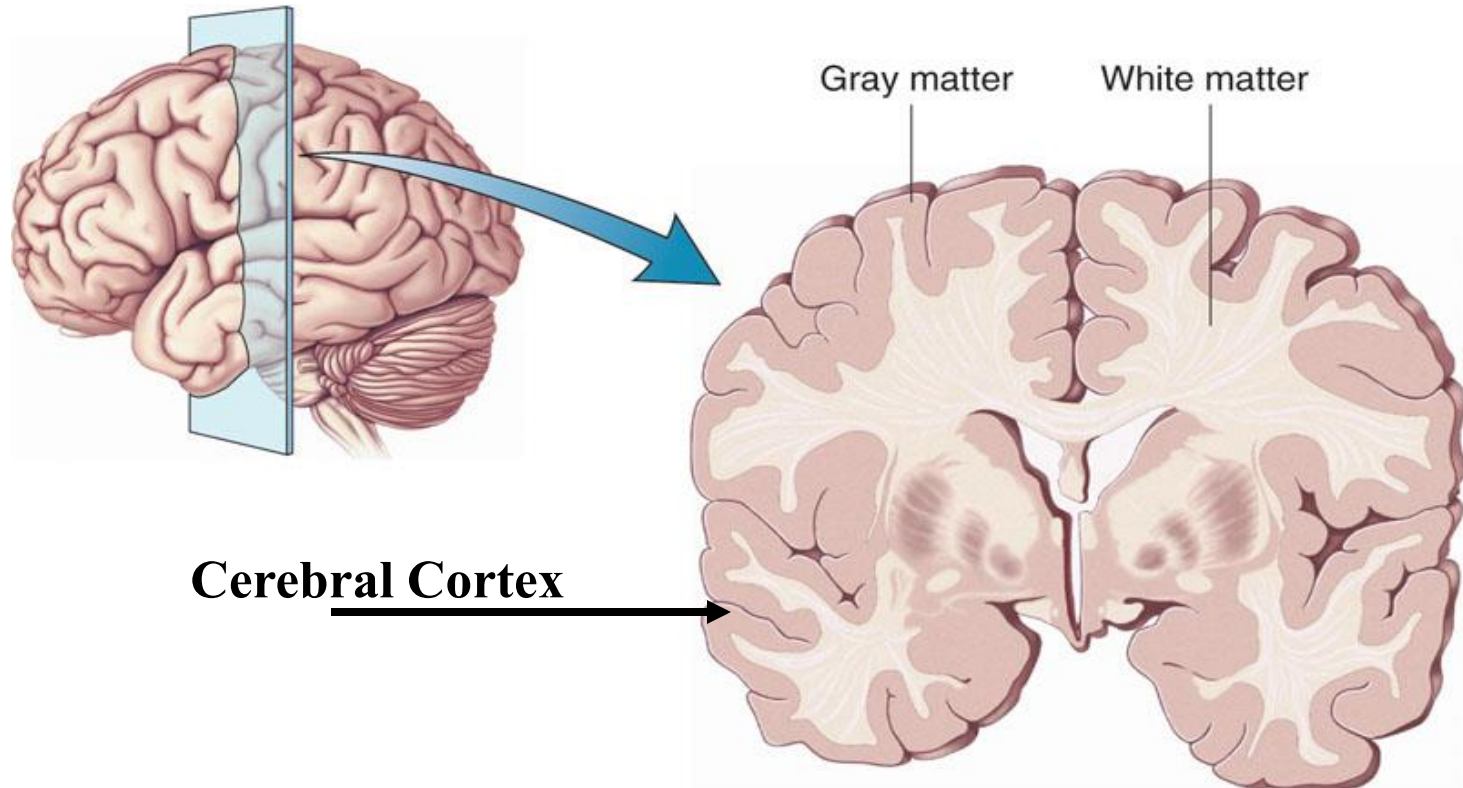
The Brain

Cerebrum -The largest division of the brain. It is divided into two hemispheres, each of which is divided into four lobes.

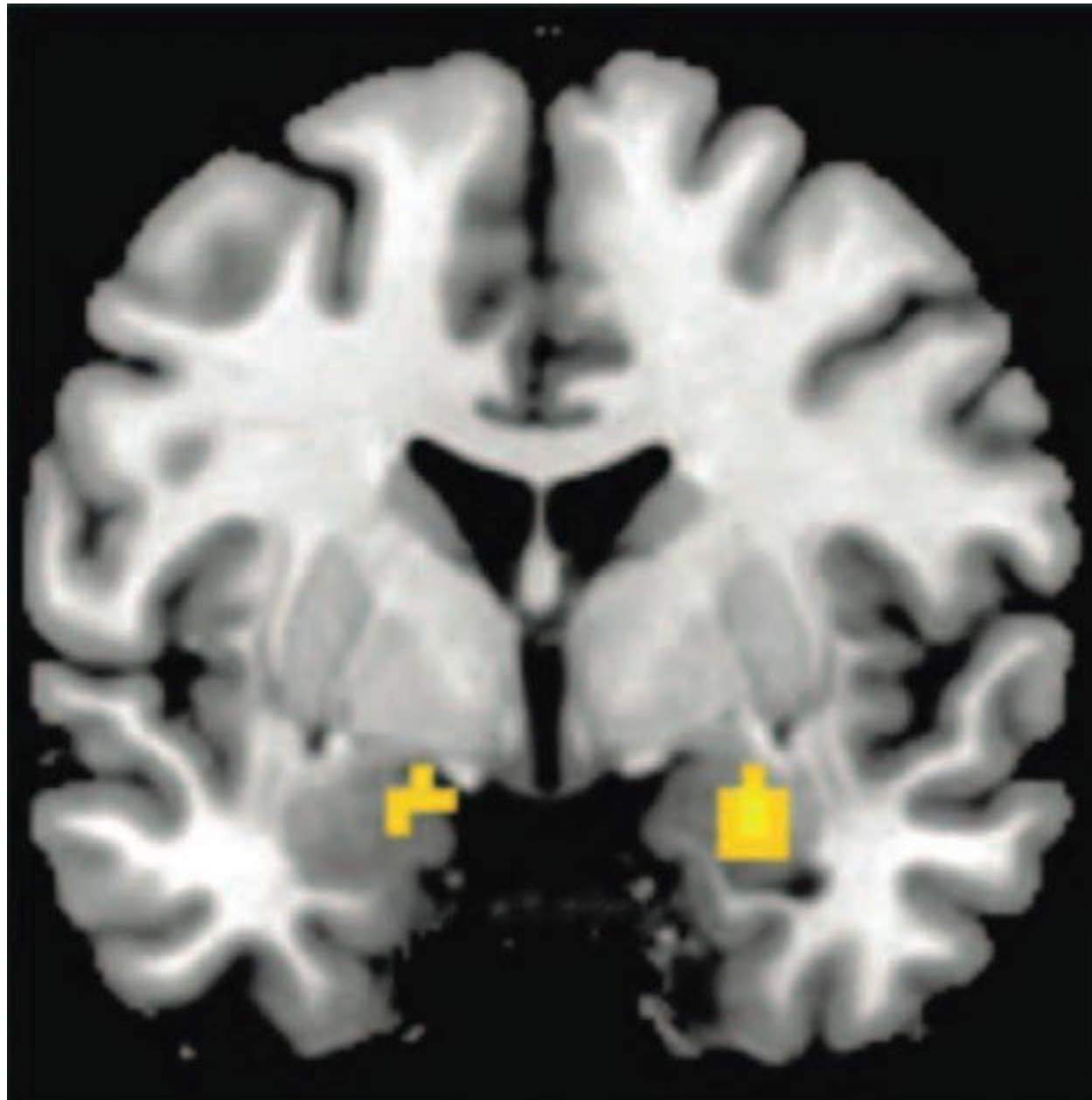


The Brain

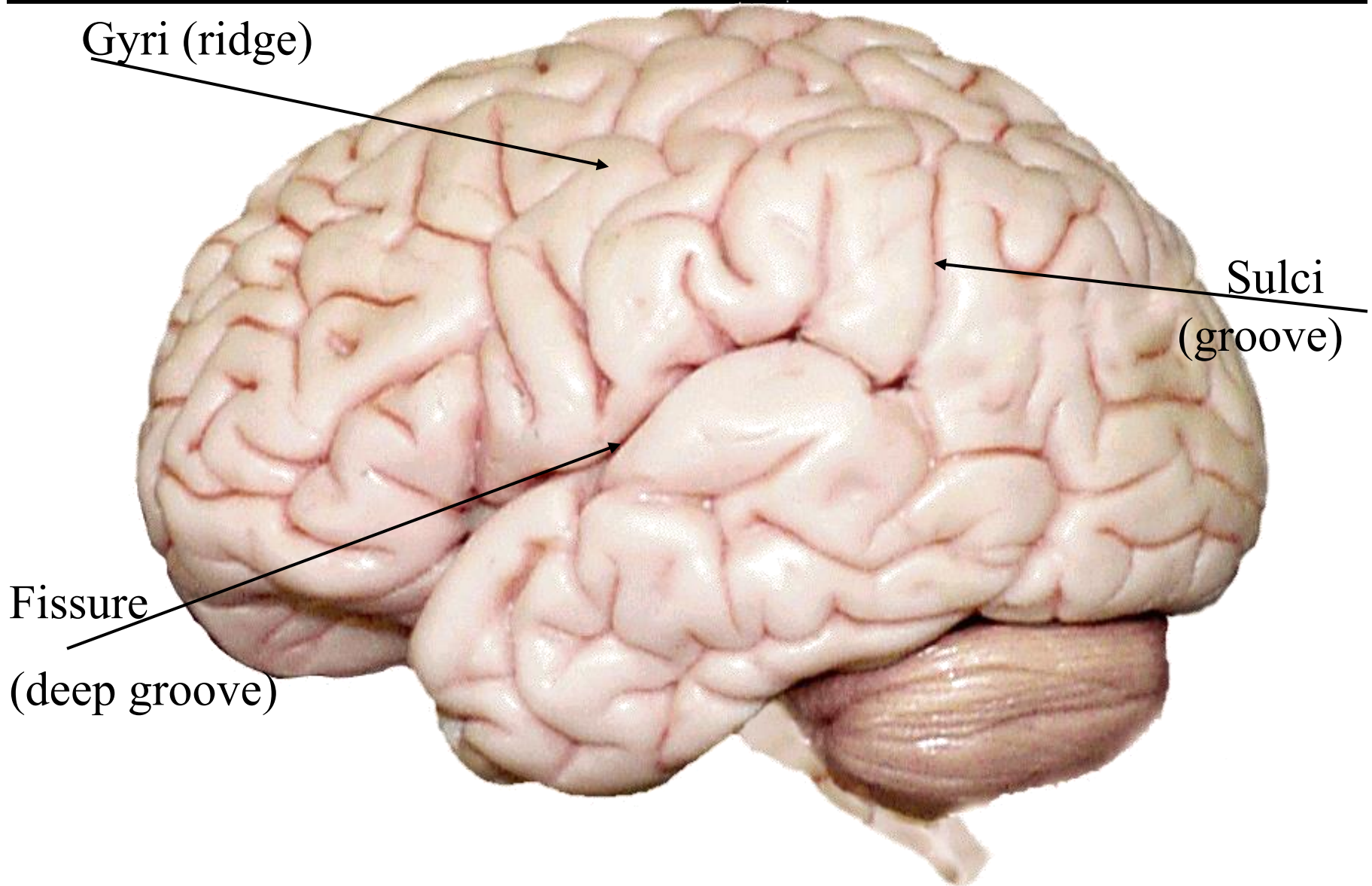
Cerebral Cortex - The outermost layer of gray matter making up the superficial aspect of the cerebrum.



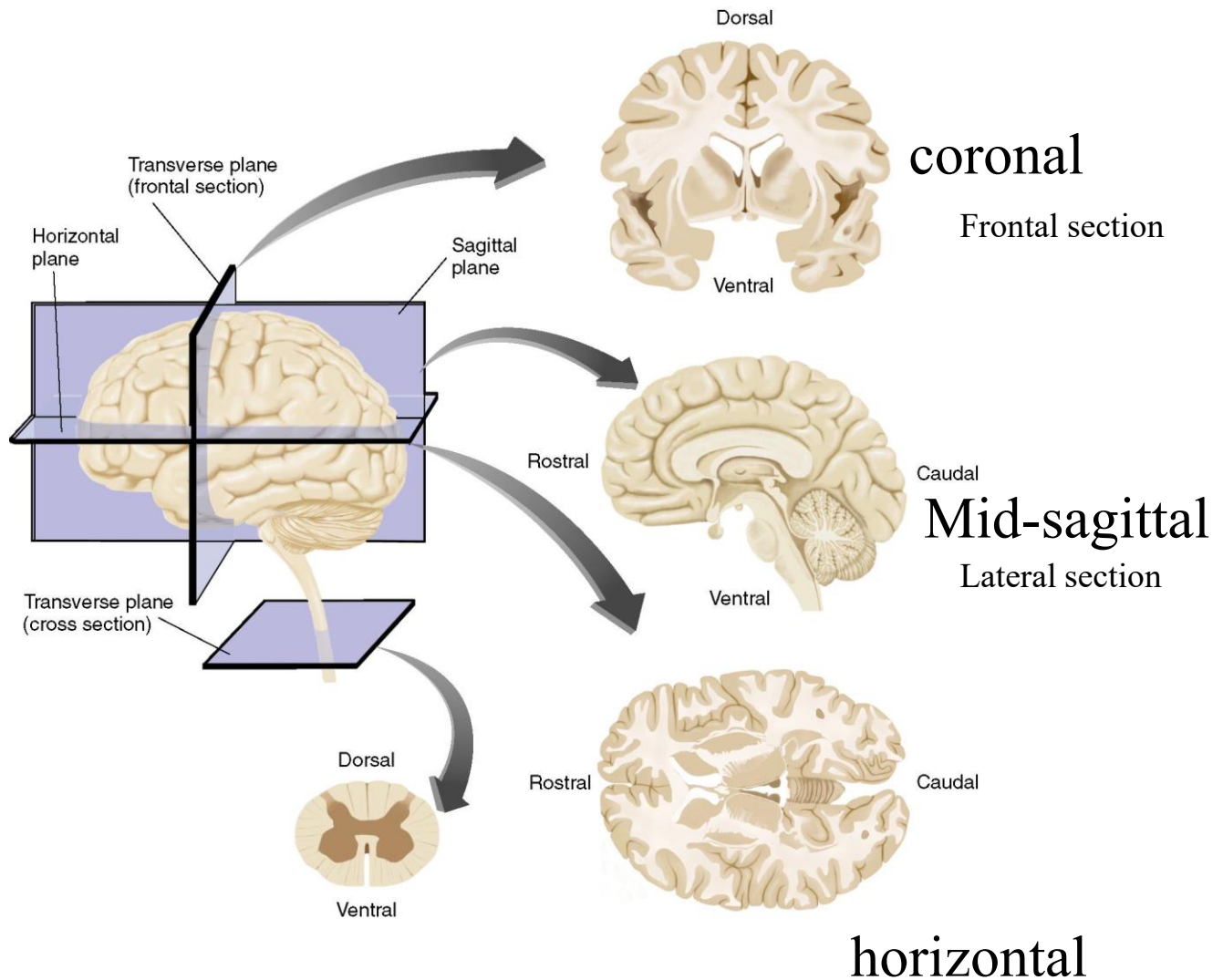
The Amygdala as an “Emotional Evaluator”



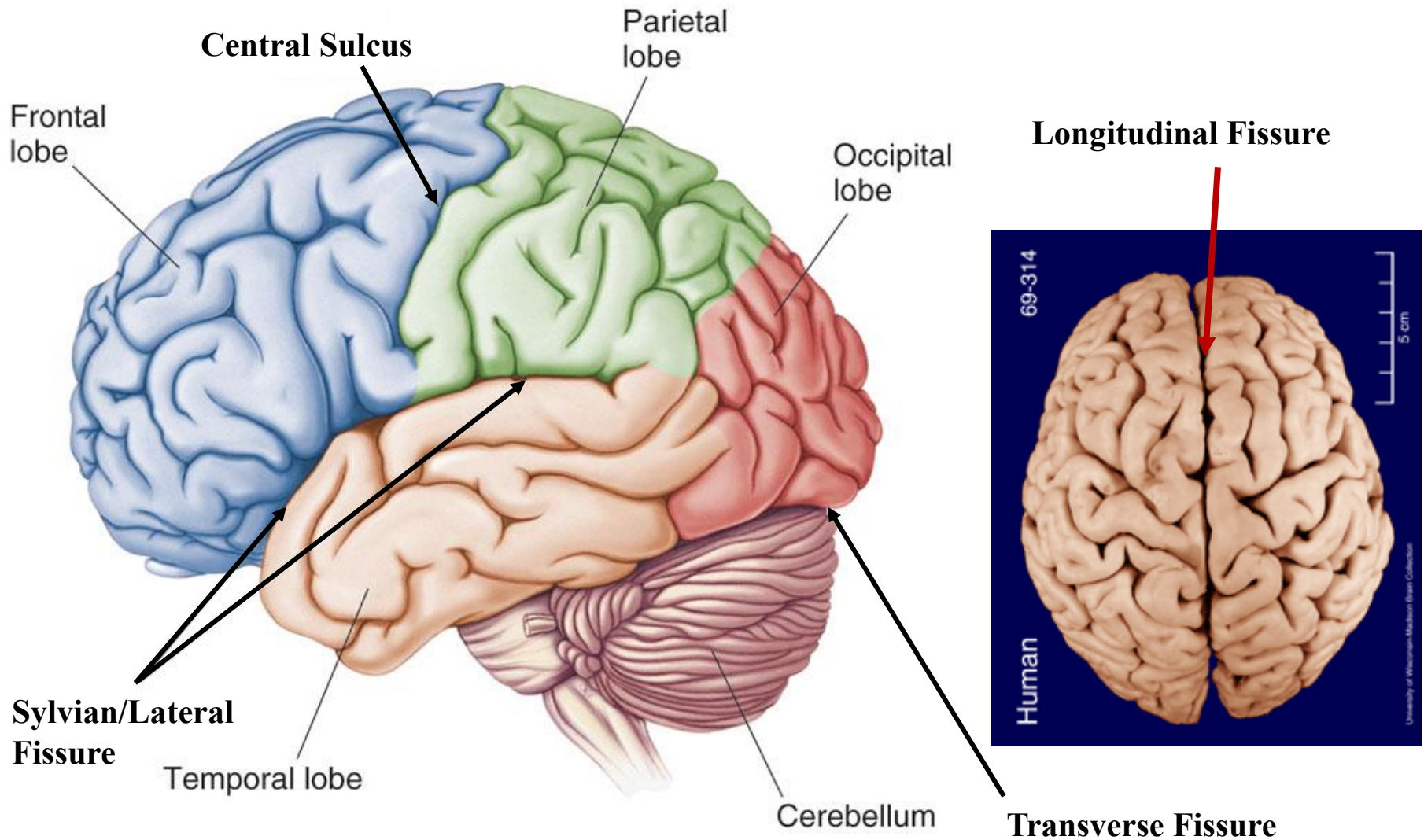
Landmarks



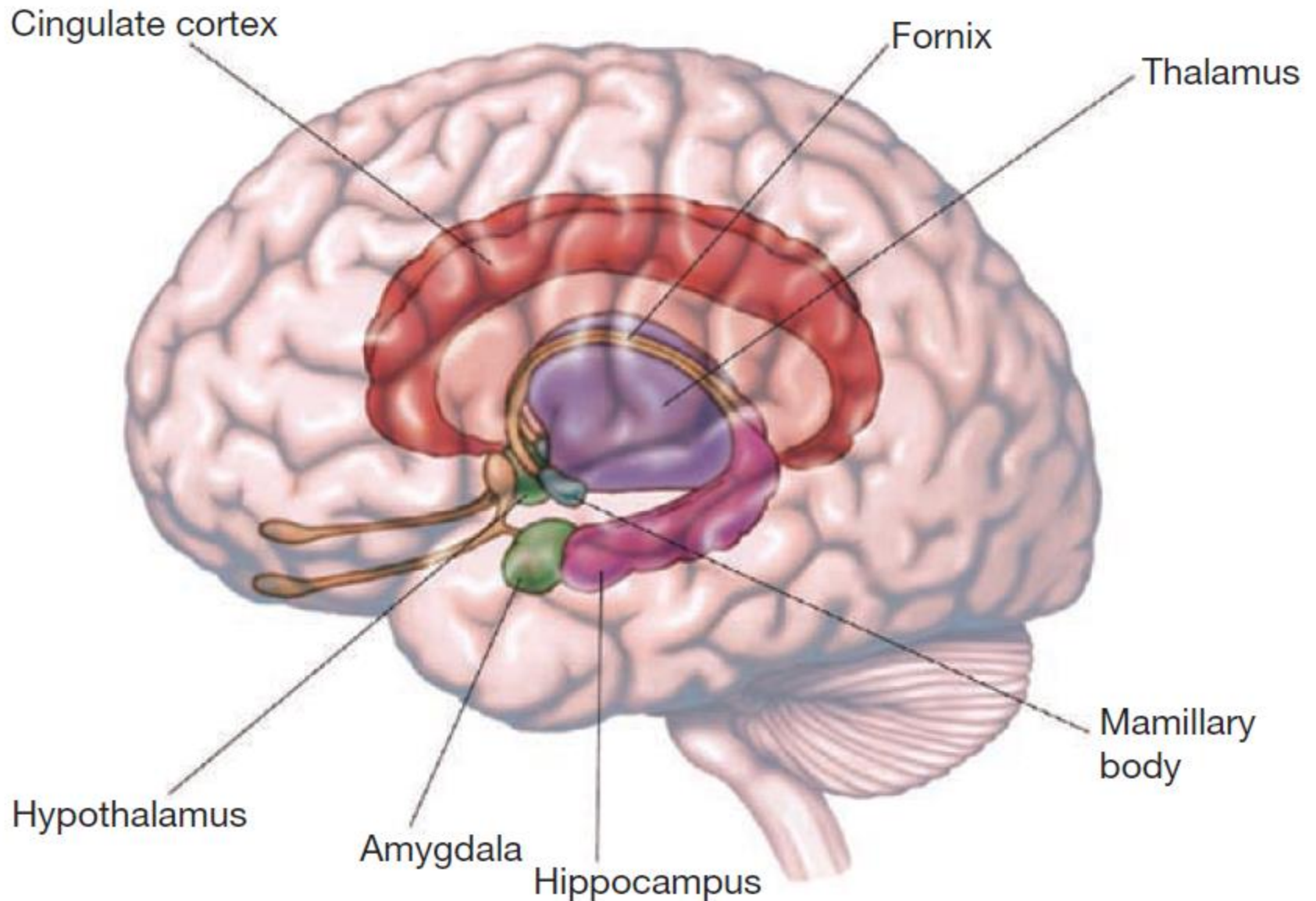
Sections of the brain



Landmarks & Lobes



Subcortical structures



Human Brain

- 2% of the body weight, 3 pounds approx.
- 25% of body's oxygen
- 70% of glucose
- Never rests and its metabolic rate in both day and night is more or less the same.
- In dreams in fact, the metabolic rate increases slightly

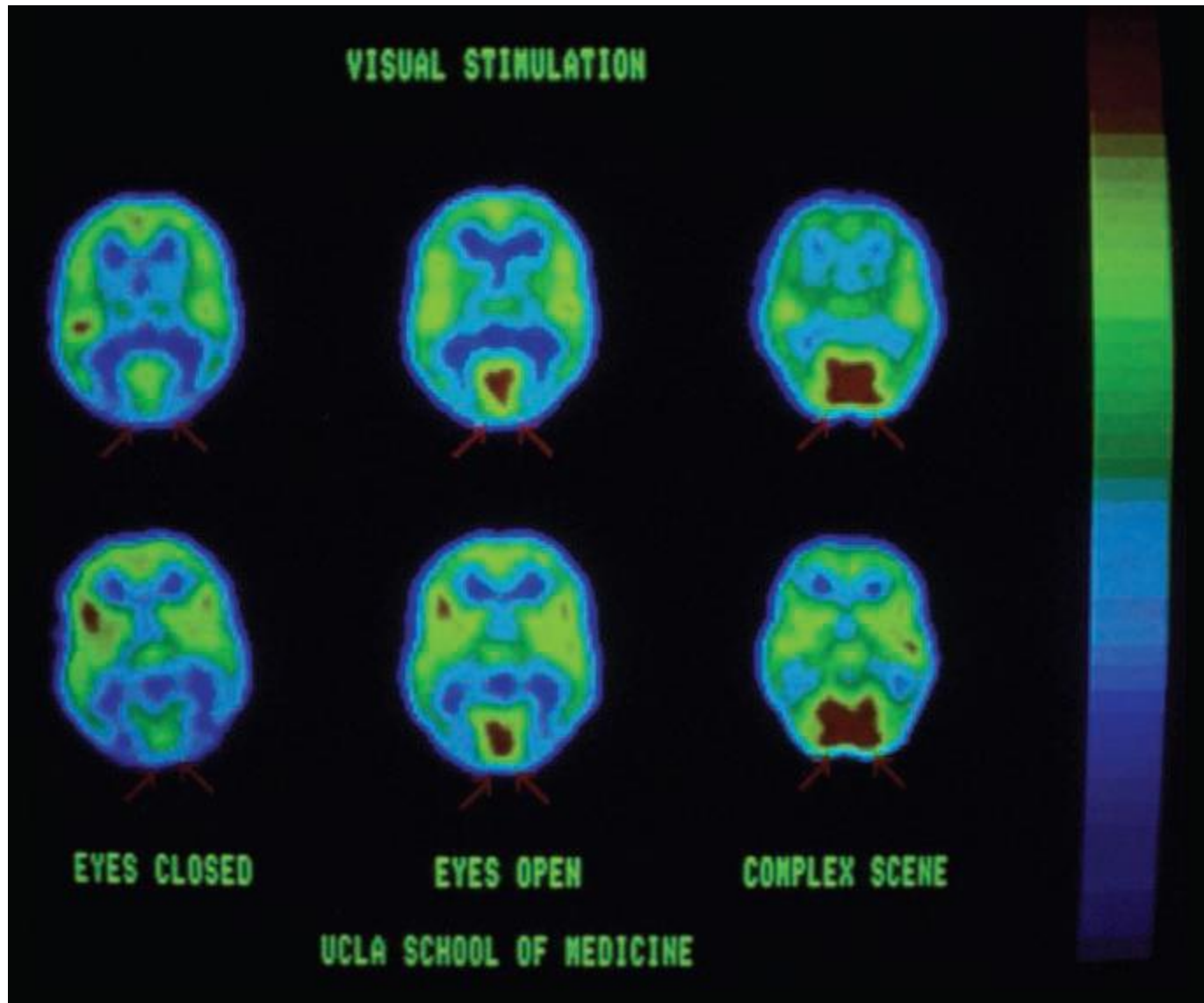
Hierarchical Brain

- Represents approx. 500 million years of evolutionary development and fine tuning.
- Core structures of brain are the same in all vertebrates
 - They govern the physiological functions
- Built upon these are newer systems that involve complex functions – sensing, emoting, thinking, reasoning etc.
- Triune brain (Paul MacLean)
 - Reptilian complex
 - Paleomammalian complex (limbic system)
 - Neomammalian complex (neocortex)

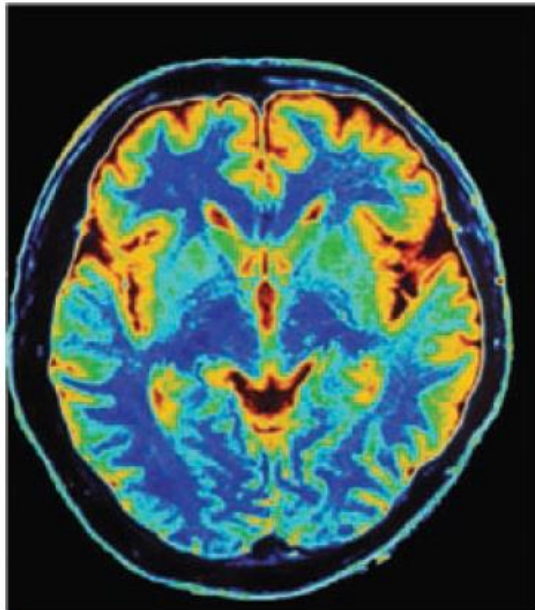
Ways of Studying the Brain

- Noninvasive:
 - MRI: Structure
 - fMRI: function
 - EEG: function
 - MEG: function
 - fNIRS: function
 - TMS: Stimulation or intervention
 - tDCS: Stimulation
- Invasive:
 - PET: Metabolic function
 - ECoG: electrode array affixed on the cortex
 - DBS: Stimulation

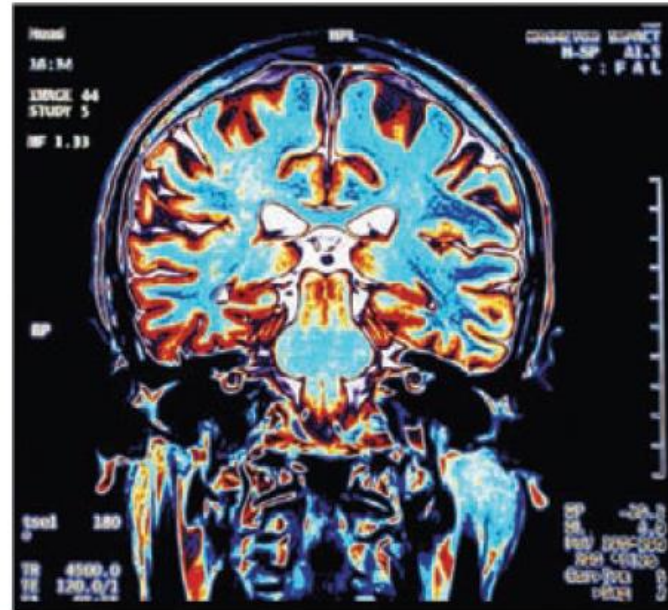
PET



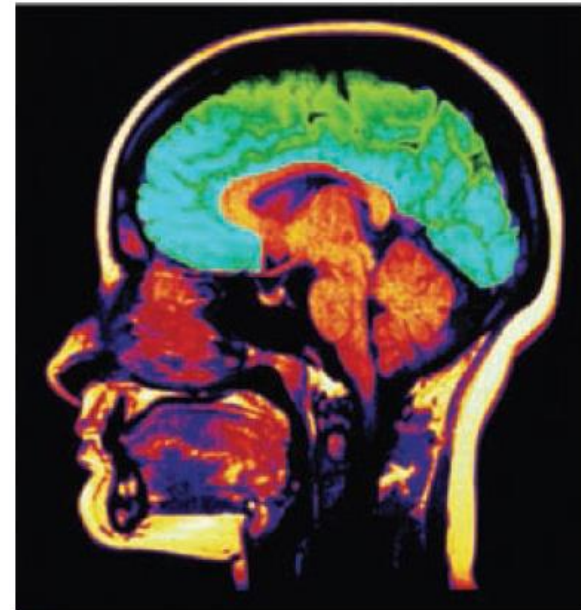
MRI (Structure)



A



B



C

EEG



Alert wakefulness



Just before sleep



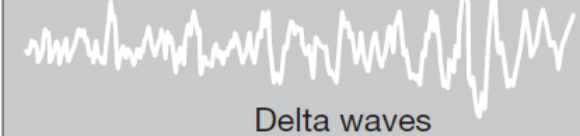
Stage 1



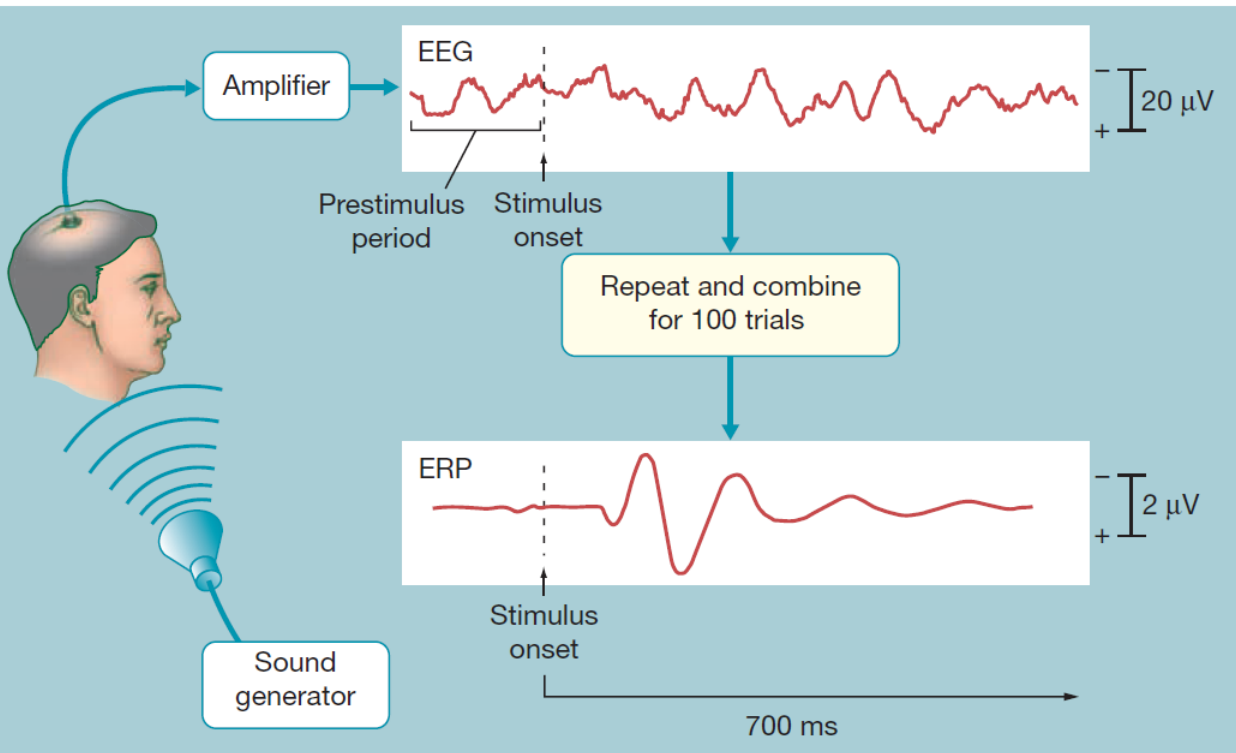
Stage 2



Stage 3

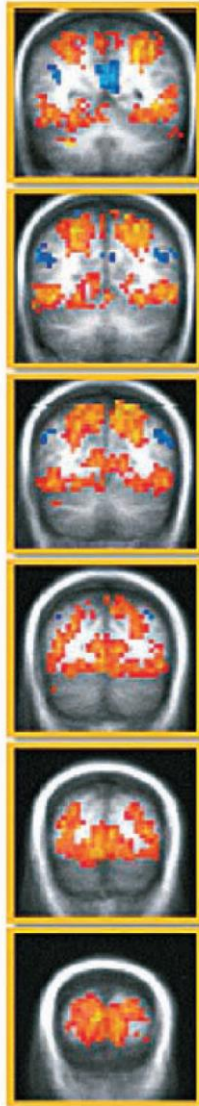


Stage 4



fMRI

Activity while
looking at
pictures



Activity while
visualizing "mental
pictures"

